Linear Algebra

Vectors, matrices, and dot products

Concept check

Name _____ Date _____

Choose one answer per question · Use the margins for calculations and sketches · 25 minutes

01 Vectors and Span

1 Why should we understand **how space changes**, as well as calculate with matrices?

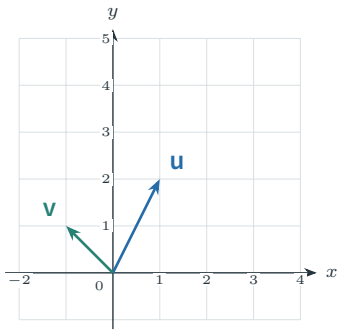
(1) A diagram always lets us skip exact numerical calculations
(2) It helps us interpret a calculation and decide which operation to use
(3) It shows that every matrix performs the same rotation
(4) Geometry is unrelated to calculation and can be memorized separately

2 **Calculate and draw $\mathbf{r} = 2\mathbf{u} - \mathbf{v}$ on the grid** Choose its coordinates

$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

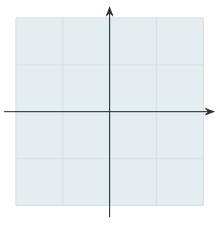
(1) $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ (2) $\begin{bmatrix} 3 \\ 5 \end{bmatrix}$

(3) $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$ (4) $\begin{bmatrix} 1 \\ 5 \end{bmatrix}$

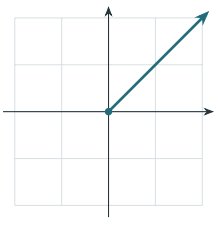


3 As a, b range over **all real numbers**, which set contains every possible vector endpoint?

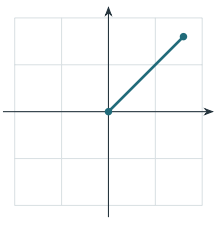
$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, $\text{Span}(\mathbf{u}, \mathbf{v}) = \{a\mathbf{u} + b\mathbf{v} \mid a, b \in \mathbb{R}\}$



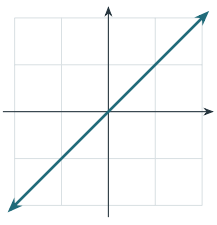
(1) The entire plane



(2) A positive ray



(3) A line segment



(4) An infinite line

02 Basis and Linear Transformations

4 Which set of vectors can be a **basis for the entire plane**?

(1) $\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}$

(2) $\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}2\\4\end{bmatrix}$

(3) $\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\0\end{bmatrix}$

(4) $\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}$

5 Which is a **linear transformation**?

$T(a\mathbf{u}+b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$

(1) $T(x,y) = (x+1,\ y)$

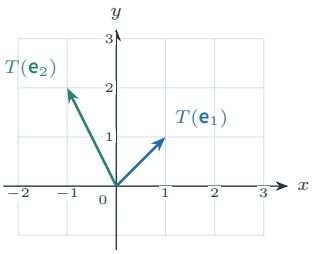
(2) $T(x,y) = (x^2,\ y)$

(3) $T(x,y) = (2x,\ x+y)$

(4) $T(x,y) = (x,\ |y|)$

6 A linear transformation T moves the two basis vectors as shown Choose **its matrix** A

$$T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}-1\\2\end{bmatrix}$$



(1) $\begin{bmatrix}1 & 1\\-1 & 2\end{bmatrix}$

(2) $\begin{bmatrix}1 & -1\\1 & 2\end{bmatrix}$

(3) $\begin{bmatrix}-1 & 1\\2 & 1\end{bmatrix}$

(4) $\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$

7 Apply T from Question 6 to $\mathbf{v}$ **Plot** $T(\mathbf{v})$ **on the grid above** and choose its coordinates

$$\mathbf{v} = \begin{bmatrix}2\\-1\end{bmatrix}, \qquad T(\mathbf{v}) = 2T(\mathbf{e}_1) - T(\mathbf{e}_2)$$

(1) $\begin{bmatrix}1\\-4\end{bmatrix}$

(2) $\begin{bmatrix}1\\0\end{bmatrix}$

(3) $\begin{bmatrix}3\\4\end{bmatrix}$

(4) $\begin{bmatrix}3\\0\end{bmatrix}$

03 Composition and Dot Products

8 Apply A **first, then** B to **x** Choose the correct combined matrix and result

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(1) BA, result $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

(2) AB, result $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

(3) BA, result $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

(4) AB, result $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\mathbf{x} \xrightarrow{A} \underbrace{\qquad}_{\text{intermediate}} \xrightarrow{B} \underbrace{\qquad}_{\text{final result}}$$

9 **Draw a perpendicular** from the tip of **w** to the line through **v** Choose the projection length and dot product

$$\mathbf{v} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \qquad \mathbf{w} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

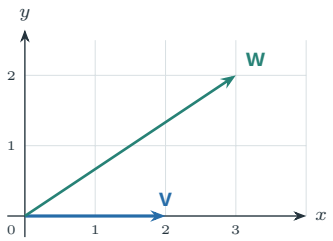
$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2$

(1) Projection length 3, dot product 3

(2) Projection length 2, dot product 4

(3) Projection length 3, dot product 6

(4) Projection length 2, dot product 6



10 A linear map L sends **a vector to one number** For $L(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x}$, choose the correct **w** and result

$$\mathbf{x} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \qquad L\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = 2, \qquad L\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = -1$$

(1) $\mathbf{w} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, result -1

(2) $\mathbf{w} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, result 5

(3) $\mathbf{w} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, result $\begin{bmatrix} 6 \\ -1 \end{bmatrix}$

(4) $\mathbf{w} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, result 7

Scope Six 3Blue1Brown videos · Questions and diagrams created separately

Visual intuition Vectors Linear combinations, span, basis Transformations and matrices Composition Dot products and duality

Content reference Official introduction transcript Vectors Span Transformations Composition Dot Products